

Project Title

Hyper-Personalized Customer Experience & Churn Prediction Engine (HPCE)

Program: *Master of Science in Data Science (MScDS)*

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1. Introduction

Customer retention is a major challenge in e-commerce. Many businesses spend a lot of effort attracting new customers, but keeping current customers is usually cheaper and more valuable over time (Gupta et al., 2006). Modern e-commerce platforms also generate large amounts of data about how customers shop and behave. This data can help businesses understand their customers better, predict future actions, and improve overall customer engagement (Ngai et al., 2009).

This project introduces the Hyper-Personalized Customer Experience & Churn Prediction Engine (HPCE), a smart analytics system that brings together machine learning, customer behavior analysis, segmentation, product recommendations, and intervention tracking. The goal is not only to find customers who may stop buying but also to suggest the right actions to keep them engaged and increase their value.

The system aims to show how an e-commerce company can move beyond simple reports and adopt a more proactive, personalized approach to understanding and supporting its customers.

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2. Problem Statement

Many customer analytics tools only show past information, such as sales, number of orders, or how many customers a business has. Even when companies use machine learning, the predictions are often not connected to real business actions (Verbeke et al., 2012). This means a business might know a customer is likely to stop buying, but they may not have a clear plan for what to do next, when to act, or how to check if the action was successful.

This project aims to fill that gap by creating a unified system that can:

- Predict which customers are at risk of leaving.
- Group customers based on their behavior.
- Automatically apply rules to choose the right personalized action.
- Recommend suitable products and strategies.
- Track how well those actions work over time.

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3. Project Aim

The goal of this project is to build a customer intelligence system that uses data to help e-commerce businesses keep their customers by predicting churn, personalizing actions, and learning from feedback.

4. Objectives

The project will focus on the following objectives:

- Build a well-organized data warehouse for e-commerce customer analytics using PostgreSQL.
- Combine transactional and behavioral data into a single customer-level analysis system.
- Create time-based and behavior-based features that show how customers act over time.
- Develop machine learning models to predict customer churn for different future periods.
- Group customers into useful behavior-based segments for targeted business actions.
- Create a rule-based system that assigns the right intervention based on customer risk and value.
- Add a recommendation module that suggests products and strategies to improve engagement and retention.
- Set up a feedback process to measure how effective the interventions are.
- Build dashboards that display churn risk, customer segments, recommendations, and intervention results.

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5. Proposed Research Questions

The project will explore the following research questions:

- How well can we predict customer churn using e-commerce purchase and behavior data?
- Which customer characteristics are most important for predicting churn?
- Can grouping customers into segments help improve retention strategies?
- How can a decision engine use both churn risk and customer segments to suggest the right business actions?
- How can we measure the results of interventions to support ongoing learning and system improvement?

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6. Significance of the Study

This project is important because it goes beyond basic dashboards and simple reports. Instead, it offers a smarter and more practical system for helping businesses retain customers. The study will be valuable for both academic research and real-world use.

From an academic point of view, it will show how machine learning, business intelligence, customer analytics, and decision systems can all work together in one framework.

From an industry point of view, it will provide a model that shows how e-commerce companies can use customer data not just to look at past behavior but also to plan future actions.

The project is also significant because it uses a feedback-based design. This means the results of each intervention can be tracked, analyzed, and used to improve the system over time.

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7. Proposed Data Sources

The main dataset for this project will be the Brazilian Olist E-Commerce Dataset, which offers realistic transactional data for customer analytics. The project will use several tables from this dataset, including:

- Customer information
- Order records
- Purchased items
- Product information
- Payment data
- Customer review data

Because transactional data does not show how customers browse or interact with the website, the project may also create a simulated event layer. This synthetic data will represent actions such as product views, adding items to the cart, and browsing sessions. Including this layer will allow the project to analyze richer customer behavior beyond just their purchases.

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8. Proposed System Architecture

The proposed system will include the following main components:

1. **Data ingestion and storage layer**
   Raw e-commerce data will be loaded into a PostgreSQL database.
2. **Feature engineering layer**
   Customer-level daily or snapshot features will be created using purchase and behavior data.
3. **Churn modeling layer**
   Machine learning models will be built to predict the probability that a customer will churn within different time periods.
4. **Segmentation layer**
   Customers will be grouped into behavior-based segments using clustering and simple rule-based methods.

5. **Decision engine layer**
   A rule-based system will suggest actions based on churn risk, customer segment, and business priorities.
6. **Recommendation layer**
   The system will provide product or engagement recommendations tailored to each customer's situation.
7. **Intervention tracking and feedback layer**
   The system will record the actions taken and analyze the results to learn what works.
8. **Visualization layer**
   Power BI dashboards will display churn risk, customer segments, recommendations, and the impact of interventions.

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9. Proposed Methodology

9.1 Data Warehousing and Preparation

The first step of the project is to load the selected Olist datasets into PostgreSQL using a dedicated schema. A structured data warehouse will be created to support analysis and modeling. The data will be cleaned, transformed, and indexed to make queries fast and to ensure good data quality.

9.2 Feature Engineering

A customer feature store will be created to capture time-based, purchase-related, and behavior-related information. The features may include:

- How recently the customer made a purchase
- How often they buy
- How much they spend
- Rolling counts of orders
- Rolling total spending
- The number of engagement events
- Days since the last event
- Total orders and revenue over the customer's lifetime
- Seasonal or time-based patterns

Feature windows such as 7, 30, 60, and 90 days may be used to capture both short-term and long-term customer behavior.

9.3 Churn Label Construction

Churn labels will be created using a rule based on future inactivity. For each snapshot date, customers who do not make any purchases within a set future period will be marked as churned. Different time windows - such as 30, 60, and 90 days - will be used so the system can support both short-term and mid-term planning.

9.4 Predictive Modeling

Several machine learning models (Hastie et al., 2009) will be tested for predicting customer churn, including:

- Logistic Regression
- Random Forest
- Gradient Boosting
- XGBoost
- LightGBM

The models will be evaluated using metrics such as:

- ROC-AUC
- Precision-Recall AUC
- Brier Score
- Business-focused ranking metrics

The project will also use time-aware validation to reduce data leakage and make the evaluation closer to real-world deployment.

9.5 Customer Segmentation

The project will also create a customer segmentation system that groups customers based on their behavior. Clustering methods like K-Means (Wedel & Kamakura, 2000) may be used with RFM-style features and engagement data.

Possible customer segments include:

- Loyal Champions
- Growth Potential
- At-Risk Loyalists
- Discount Seekers
- Dormant Customers

These segments will later help the decision engine choose personalized actions for each customer.

9.6 Decision Engine

A rule-based decision engine will be created to choose the right actions for each customer based on their churn risk, their segment, and business priorities. Customers may be grouped into risk levels such as low, medium, and high, and each level will have a matching intervention strategy.

Example actions include:

- No action
- Sending an engagement message
- Giving a loyalty offer
- Giving a retention offer
- Recommending related products (cross-sell)

9.7 Recommendation Module

A recommendation layer will be added (Ricci et al., 2011) to help personalize the customer experience. The first version will use simple and easy-to-understand methods such as:

- Popular products
- Category-based recommendations
- Suggestions from categories that help with retention
- Cross-sell opportunities

More advanced recommendation methods may be added later depending on project progress and data quality.

9.8 Intervention Outcome Tracking

To simulate a closed-loop system, the project will include a tracking method that records which actions were taken and what results followed. Measures such as response rate, orders after the intervention, revenue, and time until the next purchase will be used to see how effective each action was.

9.9 Dashboard Development

Power BI dashboards will be created to show:

- Churn risk levels
- Model results
- Customer segments
- Recommended actions

- Intervention results
- How customers move between segments over time

These dashboards will act as the main business-facing tool at the end of the project.

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10. Tools and Technologies

The project will use the following tools:

- PostgreSQL / TimescaleDB for storing data and handling time-based analysis
- Python for data cleaning, feature creation, and machine learning
- Scikit-learn, XGBoost, and LightGBM for building prediction models
- MLflow for tracking experiments
- Power BI for creating dashboards

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11. Expected Outcomes

At the end of the project, the following outcomes are expected:

1. A well-organized e-commerce customer data warehouse.
2. A customer feature store for predictive analytics.
3. Churn prediction models for different future time periods.
4. A customer segmentation system.
5. A rule-based decision engine for choosing targeted actions.
6. A basic recommendation module for personalization.
7. A framework to track and evaluate intervention results.
8. A Power BI dashboard that brings together all key insights.

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12. Expected Contributions

The proposed project is expected to make the following contributions:

- Show how churn prediction can be directly linked to business actions.
- Provide a system that combines customer segmentation, recommendations, and interventions.
- Demonstrate how feedback loops can make customer intelligence systems better over time.
- Create a practical prototype that can be used for real e-commerce retention scenarios.

13. Potential Limitations

Some limitations may arise during the project:

- The available datasets may not have complete customer behavior data.
- Synthetic events may need to be created to simulate engagement.
- Intervention results may be based on observation rather than controlled experiments.
- The quality of recommendations may depend on how detailed the product interaction data is.

These limitations will be clearly acknowledged in the final report.

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14. Proposed Timeline

A tentative timeline for the project is as follows:

Phase	Planned Activities
Phase 1	Collect data, design the database, and prepare the data
Phase 2	Create features and build churn labels
Phase 3	Develop and test the prediction models
Phase 4	Build customer segments and design the decision engine
Phase 5	Develop the recommendation module and set up intervention tracking
Phase 6	Create dashboards, run tests, and complete the final report

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15. Conclusion

This proposal outlines a plan to build a Hyper-Personalized Customer Experience & Churn Prediction Engine (HPCE) for e-commerce platforms. The system will bring together predictive analytics, customer segmentation, recommendations, and intervention tracking to create a smarter and more proactive customer intelligence solution.

If completed successfully, the project will show how businesses can move beyond basic reports and build systems that not only predict which customers may churn, but also recommend timely and personalized actions to help improve customer retention.

16. References

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